

Exp No: 24

MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

AIM:

To understand and implement a clock-driven process using:

- sc_clock
- SC_THREAD
- wait()
- Event sensitivity (positive edge trigger)

in SystemC.

PROGRAM:

```
#include <systemc.h>
```

```
SC_MODULE(ClockExample) {
```

```
    sc_in<bool> clk;
```

```
    void process() {
```

```
        while (true) {
```

```
            wait();
```

```
            std::cout << "Time: " << sc_time_stamp()
```

```
            << " - Clock Triggered" << std::endl;
```

```
        }
```

```
    }
```

```
    SC_CTOR(ClockExample) {
```

```
        SC_THREAD(process);
```

```
        sensitive << clk.pos();
```

```
    }
```

```
};
```

```
int sc_main(int argc, char* argv[]) {
```

```
    sc_clock clock("clock", 1, SC_NS);
```

```

ClockExample module("module");

module.clk(clock);

sc_start(5, SC_NS);

return 0;
}

```

